

Difference-in-Differences with Multiple Time Periods*

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Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

JEL: C14, C21, C23, J23, J38.

Keywords: Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

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These expressions for $ATT(g, t)$ clearly resemble the one for ATT in the canonical two-periods and two-groups case. As in that case, the average effect of participating in the treatment for units in group g is identified by taking the path of outcomes (i.e., the change in outcomes between the most recent period before they were affected by the treatment and the current period) actually experienced by that group and adjusting it by the path of outcomes experienced by a comparison group. Under the parallel trends assumption, this latter path is the path of outcomes that units in group g would have experienced if they had not participated in the treatment.

Remark 3. From (2.8) one can see that when Assumptions 3 and 4 hold unconditionally and there is no-anticipation, the $ATT(g, t)$ parameter can be obtained by first subsetting the data to only contain observations at time t and $g - 1$, from units with either $G_g = 1$ or $C = 1$, and then, using only the observations of this subset, running the (population) linear regression

$$Y = \alpha_1^{g,t} + \alpha_2^{g,t} \cdot G_g + \alpha_3^{g,t} \cdot 1\{T = t\} + \beta^{g,t} \cdot (G_g \times 1\{T = t\}) + \epsilon^{g,t}. \quad (2.10)$$

It is then easy to verify that $\beta^{g,t} = ATT(g, t)$. Note that one would need to consider different partitions of the data to characterize different $ATT(g, t)$ in terms of regression parameters. Alternatively, one could use the interacted two-way fixed effects regression proposed by [Sun and Abraham \(2020\)](#).

Remark 4. When covariates are available, the $\tilde{\beta}^{g,t}$ coefficient of the population linear regression

$$Y = \tilde{\alpha}_1^{g,t} + \tilde{\alpha}_2^{g,t} \cdot G_g + \tilde{\alpha}_3^{g,t} \cdot 1\{T = t\} + \tilde{\beta}^{g,t} \cdot (G_g \times 1\{T = t\}) + \tilde{\gamma} \cdot X + \tilde{\epsilon}^{g,t}$$

that uses the same subset of data as in Remark 3 is, in general, not equal to $ATT(g, t)$ unless one is willing to assume (i) homogeneous (in X) treatment effects, i.e., $\mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1, X] = \mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1]$ a.s., and (ii) rule-out covariate-specific trends, i.e., for $\mathbb{E}[Y_t - Y_{t-1}|X, G] = \mathbb{E}[Y_t - Y_{t-1}|G]$ a.s. for all groups and time periods; see, e.g., [Śłoczyński \(2018\)](#) for a related discussion. The characterizations of $ATT(g, t)$ discussed in Theorem 1 do not rely on these restrictive conditions.

Remark 5. Although the IPW, OR, and DR based estimands presented in Theorem 1 are identical from an identification standpoint, this is not the case when one wants to estimate and make inference about the $ATT(g, t)$. As we discuss in Section 4, DiD estimators based on the DR estimands (2.4) and (2.7) usually enjoy additional robustness against model-misspecifications when compared to the IPW and OR estimands.

Remark 6. Theorem 1 suggests that we can identify $ATT(g, t)$ only for groups in $\mathcal{G}_\delta \subseteq \mathcal{G}$ which can involve dropping some “early treated” groups due to anticipation effects. When $\delta = 0$, i.e. when there is no anticipation, $\mathcal{G}_\delta = \mathcal{G}$. Theorem 1 also suggests that we can identify $ATT(g, t)$ only until $t = \mathcal{T} - \delta$ because of potential treatment anticipation behavior. In applications where some units are known to never participate in the treatment (including periods after time period $\mathcal{T}$), however, we can identify $ATT(g, t)$ up to $t = \mathcal{T}$ by using these units as a valid comparison group for all time periods $t = \mathcal{T} - \delta + 1, \dots, \mathcal{T}$, provided that an appropriate parallel trends assumption is satisfied.

Remark 7. From Theorem 1 it is clear that pre-treatment covariates play a prominent role in our analysis. Importantly, Assumptions 4 and 5 suggest that researchers should include pre-treatment covariates